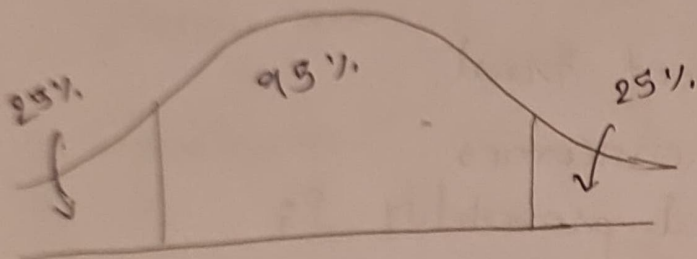


Confidence Interval

ϕ
Margin of Error.

C.I. = 95%.



Point Estimate:

$\bar{x}$

$\bar{x}$
 $\bar{x} = 2.5$

μ
 $\mu = 3$

S.S.

It might possible that sample mean is greater as well as less than population mean.

Confidence Interval:

Point Estimate $\pm$ Margin of Error.

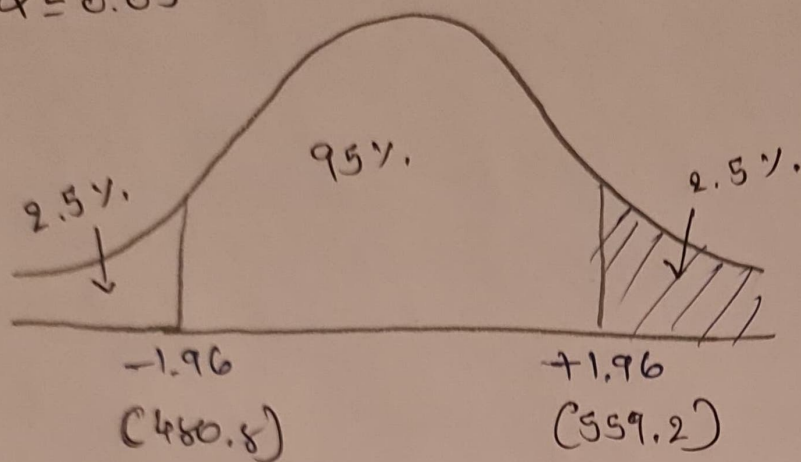
If Z test $\Rightarrow$
$$\bar{x} \pm Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

If t test then at the place of Z use t & also find degree of freedom too.

Ex/ On the verbal section of CAT exam, the standard deviation is known to be 100. A sample of 30 test takers has a mean of 520. Construct 95% C.I. about mean.

$$\alpha = 0.05$$

$$\bar{x} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}}$$



$$1 - 0.025 = \underline{\underline{0.9750}}$$

+1.96 $\Rightarrow$ from Z-table

$$\text{Lower C.I.} = 520 - (1.96) \frac{100}{\sqrt{25}} = \underline{\underline{480.8}}$$

$$\text{Higher C.I.} = 520 + (1.96) \frac{100}{\sqrt{25}} = \underline{\underline{559.2}}$$

Conclusion: I am 95% confident about the mean CAT score is between 480.8 and 559.2.